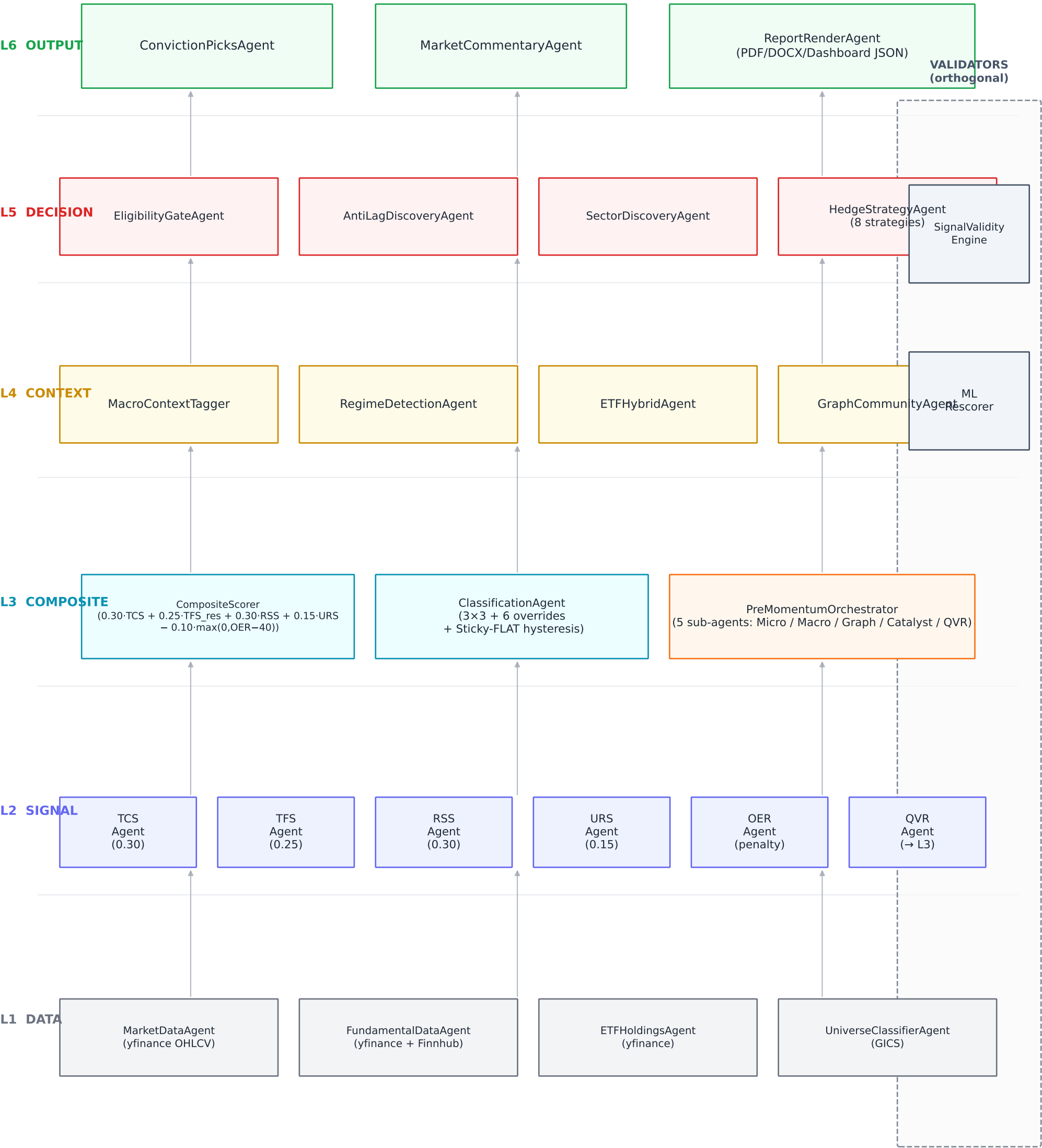


Price Discovery — Agent Architecture

23 in-house agents + LLM agent layer (Claude Max plan)



See PAGES 2-6 for per-tier detail. LLM layer (PAGE 6) is fed exclusively from within Claude Code sessions — no out-of-plan API spend.

Per-ticker, vectorized compute → cross-sectional standardization

L1 — Data Layer (4 agents)

MarketDataAgent

- yfinance OHLCV
- adj-close, fast_info
- 770 tickers × T days

FundamentalDataAgent

- yfinance + Finnhub
- PE / PEG / margin / ROE
- revisions, surprises

ETFHoldingsAgent

- yfinance top_holdings
- 70 sector / thematic ETFs
- weekly refresh

UniverseClassifier

- GICS sector / industry
- cap-tier / region

L2 — Signal Agents (6, per-ticker)

Each agent follows compute_raw() → score() → percentile_rank(). All outputs are 0..100 and are cross-sectionally standardized before flowing into L3.

TCSAgent

Trend Continuation

- SMA20/50/200 distance + slope
- trend age (long-horizon weighted 60%)
- score_tcs_short + score_tcs_long

TFSAgent

Trend Formation (residualized)

- volatility compression, BB squeeze
- volume surge, structural breakout
- cross-sectional OLS removes TCS overlap

RSSAgent

Relative Strength (hybrid)

- 0.6 within-sector + 0.4 universe
- rss_short / rss_long percentiles
- small categories n<8 → universe fallback

URSAgent

Underreaction

- LeadLag + AttnGap + Drift + Dispersion
- behavioral overlay (post-event)
- 4 sub-signal blend

OERAgent

Overextension (penalty)

- 52w-high distance, RSI(14), dist_days
- feeds Composite as −0.10·max(0, OER−40)
- asymmetric: caps at −6pts

QVRAgent

Quality + Value + Revision

- cross-sectional percentile of Q/V/R
- doubles as Eligibility Gate input
- Stock-only; ETFs bypass QVR ≥ 40 check

Cross-sectional standardization happens ONCE per scan (single pass): TFS residualized vs TCS, RSS hybrid blend, universe percentiles. Output is consumed by L3 CompositeScorer.

Where signals become decisions: 4-axis composite + 3×3 matrix + 5-agent PM

CompositeScorer

Formula (2026-05 update)

base = 0.30·TCS + 0.25·TFS_residualized + 0.30·RSS_hybrid + 0.15·URS
Composite = base − 0.10·max(0, OER − 40) # OER penalty, capped −6 pts

- TFS_residualized via cross-sectional OLS (removes TCS↔TFS info overlap)
- RSS_hybrid = 0.6 · within-sector percentile + 0.4 · universe percentile
- OER < 40 → no penalty; OER = 100 → −6 pts (penalizes overheated names, not healthy trends)

ClassificationAgent — 3×3 Matrix

← long_dir →

	UP	FLAT	DOWN
UP	CONT.	RECOV.	COUNTER
FLAT	CONSOL	NEUT.	FADING
DOWN	PULLBK	WEAKEN	DOWN

Overrides (6) + Hysteresis

OVEREXTENDED — OER ≥ 60 on bullish base
FORMATION — rapid short breakout
EXHAUSTING — peak signs after long uptrend
CYCLE_PEAK — top of full cycle (regime-aware)
LAGGING_CATCHUP — catching up to peers
→ Phase 3B: regime-aware override in Risk-Off ×
 cyclical × OVEREXTENDED × OER≥70 → CYCLE_PEAK

STICKY-FLAT HYSTERESIS
if prev_classification = FLAT → entry threshold × 1.3
→ suppresses bi-weekly NEUTRAL↔CONSOL↔RECOV flips

PreMomentumOrchestrator — 5 sub-agents

Microstructure 0.20	MacroRegime 0.15	GraphRelational 0.20	Catalyst 0.20	QVR 0.25
vol compression accumulation BB squeeze	6 sub-signals → rotation_alignment → etf_parent_signal	theme breadth leader-lagger gap community momentum	rss_short − rss_long strategy_agreement reversal_risk	★ only PM agent orthogonal to Composite (corr ≈ 0)

Cross-sectional macro overlay + multi-tier eligibility logic

L4 — Context & Regime (4 agents)

MacroContextTagger

- Per-ticker tags (Phase 1.0/1.5):
- cyclical_tag : cyclical / defensive / broad
 - style_tilt : growth / value / balanced
 - region : US / Korea / Japan / China / ...
 - industry_group: GICS industry group name

Industry refinement:
Biotech → cyclical/growth, Telecom → defensive/value

RegimeDetectionAgent

- STATE['regime']:
- cyclical_dom / defensive_dom (gap > 3)
 - growth_dom / value_dom
 - top_region / bot_region
- Per-ticker:
- rotation_long / rotation_short (0..100)
- Phase 2C alignment, Phase 3B CYCLE_PEAK

ETFHybridAgent

- 7-field sidecar per ETF:
- constituent_breadth_mom / weighted_comp
 - coverage / concentration / leader_gap
 - parent_etf_signal (for stocks)
- 4 divergence flags:
HEALTHY_TREND / NARROW_RALLY / STEALTH_STRENGTH / WRAPPER_DRAG

GraphCommunityAgent

- GraphRAG knowledge graph (graph_engine.py):
- Louvain communities (theme clusters)
 - theme propagation insights
 - leader-lagger detection
 - ETF-stock divergence flags
 - multi-hop: impact_radius / theme_status

L5 — Decision & Filtering (4 agents → tiers)

EligibilityGateAgent

- 4 conditions, ALL required:
- Composite ≥ 55
 - classification ∈ bullish set
 - ADV ≥ \$5M (liquidity floor)
 - asset = ETF OR QVR ≥ 40
- Pass → Momentum tab | Fail → Excluded tab (with rejection tag)

AntiLagDiscoveryAgent (Phase 1)

- ProvisionalPM tier:
PM ≥ 45 AND agreement_ratio ≥ 0.6 AND
bullish classification AND NOT eligible
- surfaces strong forward-looking signals BEFORE
Composite breakout. Goal: 10-15-day lag reduction.
→ Anti Lag Discovery sub-tab

SectorDiscoveryAgent

- Per-sector top-5 (forced diversification):
within each sector, top-5 by Composite
AND composite ≥ 40 AND bullish
- Sector Discovery sub-tab
Goal: forced diversification + sector-best capture
(does NOT reduce lag)

HedgeStrategyAgent

- 8 quant strategies (hedge_strategies.py):
Combined / Flow / Wyckoff / Ichimoku /
Darvas / Minervini / sector_rotation / ...
- Per-strategy long/short top picks → surfaced via
/api/quant-strategies, Strategy cards in dashboard

Top of stack: synthesis + orthogonal hit-rate validation

L6 — Output & Synthesis (3 agents)

ConvictionPicksAgent

Buy/Sell top-5 Stocks + ETFs:

BuyScore = Composite + Class + Consensus + 1M/Sector regime

- OER penalty
- WeakQVR penalty (Stock-only)
- + ETF flag bonus (HEALTHY_TREND etc.)

(!) amber row highlight for WeakQVR Buy candidates

MarketCommentaryAgent

~15k-char auto-report:

Executive Summary + Conviction Picks +
Market Leaders (5-axis leadership) +
ETF Hybrid Health + 23 narrative sections

Includes the new ConvictionDebateCard panel
(L6 ↔ LLM layer integration — see PAGE 6)

ReportRenderAgent

Multi-format output:

- PDF — daily Omega(PD_v5)_YYYYMMDD.pdf in reports/ (matplotlib dark-theme)
- DOCX — earnings-analysis skill outputs (e.g. Nvidia_Q1_FY27_Earnings_Update.docx)
- FastAPI JSON — /api/table, /api/pre-momentum, /api/conviction-debate, /api/sector-rotation, ...
- React dashboard (frontend/) — 8 top-level tabs, 5 PD sub-tabs, ConvictionDebateCard
- Sector overview reports (markdown) — pitch-agent:sector-overview outputs

Orthogonal Validators (SVE + ML)

SignalValidityEngine

24 bi-weekly Friday-anchored eval points × 63d:

- bucket hit rates (Composite quintiles)
- per-class hit rates
- per-ticker hit rates
- segmented by:
 - cyclical_tag (cyclical/defensive/broad)
 - style_tilt (growth/value/balanced)
 - region
 - llm_catalyst_event_type (new)

Feeds /api/validation + Validation tab

MLRescorerAgent

ML overlay on top of L3 outputs:

- feature_pipeline.py — engineered features
- score_ml.py — LightGBM rescorer
- purged_cv.py — leakage-safe CV
- meta_labeling.py — tri-class meta labels

Output: /api/ml/* mirror endpoints,
Price Discovery (ML) tab + AI Prediction tab

Used for the composite_live + ml variants of
sector_rotation backtest (signal_mode params)

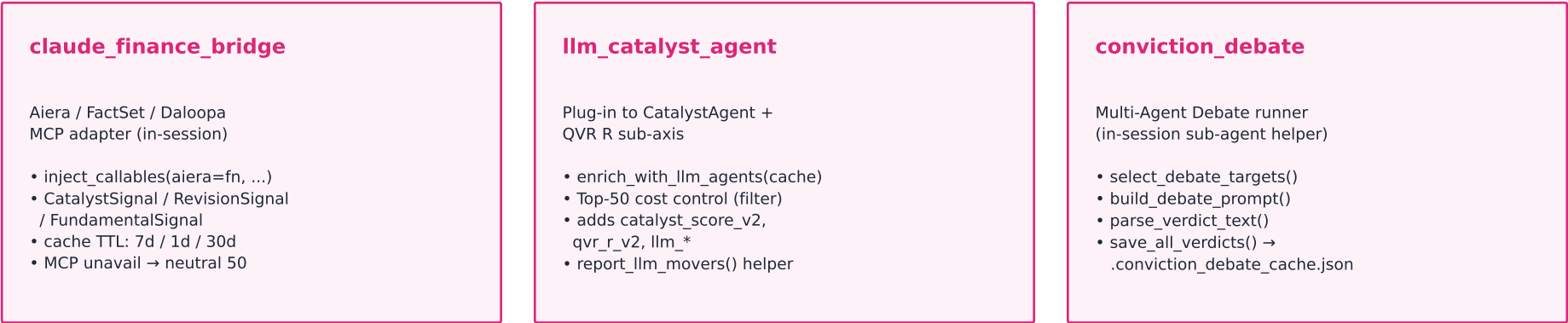
- DAILY EXECUTION FLOW:
1. price_discovery.py (full scan, ~15min) → .scan_cache.pkl
 2. fundamentals_pipeline.py → .fundamentals_cache.pkl
 3. finnhub_fundamentals.py (US enrich, ~28min)
 4. unicorn restart → api.py post-load applies L4 + L5 + Pre-Mom
 5. SVE runs inside run_scan() (no separate pass)
 6. MLRescorer runs in /api/ml/* endpoints (lazy)

→ New layer (PAGE 6): in-session LLM enrichment via Claude Code (Manual or CronCreate)

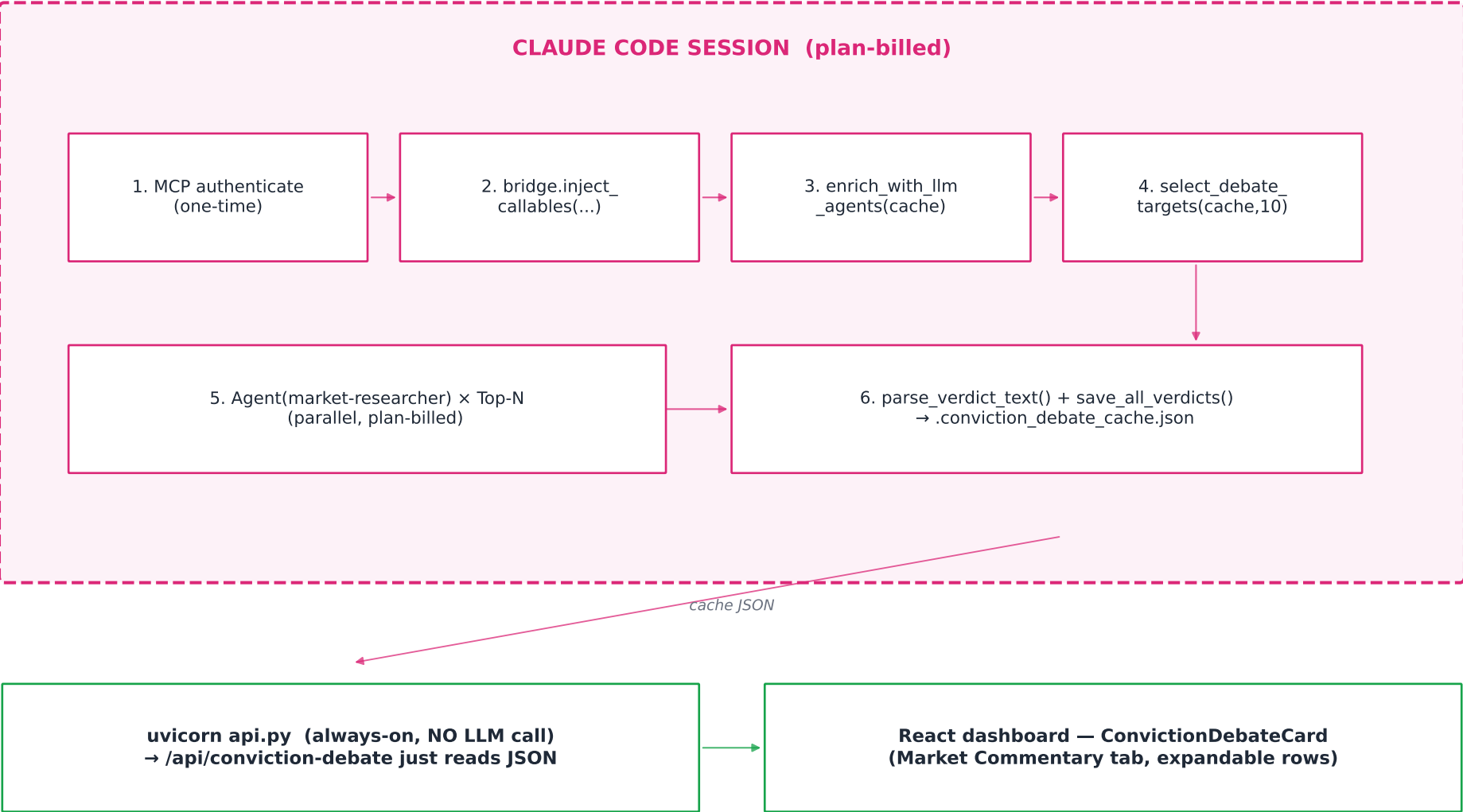
~~External MCP data + Multi-Agent ConvictionDebate, no out-of-plan API spend~~

HARD RULE — No code in agents/ ever calls the Anthropic API directly. All LLM / sub-agent / MCP work happens inside an active Claude Code session.

Three pure-Python agents (no network I/O)



Execution Flow — Claude Code session boundary



Two run modes (both inside Claude Max plan)

